

1-Deck Card → BIP-39 — how it works

Generator v2.15 · derivation id “1deck-bip39-v2” · one page. Full spec: techspecs.md

WHAT IT DOES

You shuffle a real 52-card deck, draw four cards at a time and type them in. Those cards, and nothing else, become a BIP-39 seed phrase. There is no random number generator anywhere in the file — no `Math.random`, no `crypto.getRandomValues`, not even a clock reading. If your shuffle is good, your seed is good; if it is not, no software can rescue it.

THE DERIVATION — THE WHOLE THING

```
preimage = "1deck-bip39-v2|12|TD KD 8H 5C TS 2D 8C JH ..."  
entropy  = first 128 bits of SHA-256(preimage)      (256 bits for 24 words)  
checksum = first 4 bits of SHA-256(entropy)         (8 bits for 24 words)  
mnemonic = entropy+checksum, split into 11-bit groups, indexing the BIP-39 list
```

Cards are written rank then suit, uppercase, T for ten, single spaces, in the order dealt. Order matters: the same four cards dealt in a different order give a different seed. The derivation id is frozen — it never tracks the version number, so every 2.x build rebuilds the same seed from the same cards.

HOW MANY DRAWS, AND WHY

```
12 words -> 8 draws of 4 cards = 32 entries    181 bits of card entropy -> 128 used    +53 spare  
24 words -> 14 draws of 4 cards = 56 entries    317 bits of card entropy -> 256 used    +61 spare
```

You need one 52-card deck, not 56 cards. The deck is reshuffled before every draw, so the same card can come up again. 56 is how many card entries you write down — not how many cards you have to own.

An ordered draw of four cards from 52 is worth $\log_2(52 \times 51 \times 50 \times 49) = 22.63$ bits. We deliberately collect far more than we use. That surplus is the whole point: SHA-256 only behaves as a proper randomness extractor when its input carries more entropy than its output. Feed it exactly as many bits as you take out and an imperfect shuffle passes straight through.

SHUFFLE PROPERLY — THIS IS THE WEAK LINK

- Wash first: cards face down on the table, swirled with both hands for 30 seconds or more. This destroys the factory order far faster than riffing.
- Then riffle at least seven times, with a cut or strip between each. Measured: after three riffles a deck still shows about three times as many adjacent same-suit pairs as a random one. Overhand or Hindu needs roughly 50 passes, not five.
- Use a sealed deck bought new. Check for drilled holes or clipped corners — a cancelled casino deck can arrive in a known order and no software check would notice.
- The built-in shuffle check is a blunder detector, not a certificate. It catches one or two riffles reliably, is blind past about four, assumes the deck started in factory order, and a deterministic faro shuffle scores better than random.

VERIFY BEFORE YOU FUND ANYTHING

- Use verify-by-re-entry: the tool blanks the grid, you retype your draws from paper, and it confirms the two entries match without ever showing the first back to you.
- Restore into your wallet and compare the master fingerprint — eight hex characters — and the first two receive addresses. If they match, every word arrived intact.
- Send a small amount first. Recover it. Only then commit real funds.

IF THIS TOOL EVER DISAPPEARS

Your card record is the backup. The entropy hex printed in the PDF seed record is the universal interchange format — every BIP-39 implementation accepts it.

```
printf '%s' '1deck-bip39-v2|12|<your cards>' | sha256sum -> first 32 hex chars
```

Paste that into any BIP-39 tool with entropy type set explicitly to Hex — do not rely on auto-detection. On iancoleman, leave Mnemonic Length on “Use Raw Entropy”; anything else hashes your entropy again and gives a different seed. Do not paste the cards into another tool and expect these words: other tools encode cards differently. The entropy hex is what they all share. `recover.py`, in the repository, does the whole job with Python's standard library, no network.

A WORKED EXAMPLE YOU CAN CHECK YOURSELF

```
cards      TD KD 8H 5C   TS 2D 8C JH   8C 9C KH TC   KD 7H 3S QH  
           8H 7H AD JH   7S KS 5S 6C   5D QH 7S 3C   AC JS QS 3H  
entropy    56cd7feb1647fd133477bc54589f7c2d  
phrase     fine hip width clutch lemon maze spike wasp february shaft tenant force  
XFP        B8428EFE           first address bc1qv96hhcapy3x55jk8w59nsxeut5vytnuga8fn6l
```

These cards are published. Never put funds on this seed — it is a test vector, not a wallet.

THE PAPER AND THE PDF ARE YOUR SEED

The card draws alone reproduce the seed. Redacting the words does not make either safe. Guard them accordingly.